

Is Vitamin E Involved in the Autoimmune Mechanism?

Samuel Ayres, Jr, MD, FACP, Los Angeles, California

Richard Mihan, MD, Los Angeles, California

Autoimmune diseases are characterized by an alteration of the body's defense mechanism, designed for protection against infections and toxic injuries, which for unknown reasons attacks and destroys normal tissue.

Some evidence strongly suggests that such diseases are the result of hydrolytic enzymes that escape from lysosomes whose membranes have been damaged by lipid peroxidation or other causes and that combine with and denature normal tissue proteins—in effect converting them into foreign proteins—to which the body then reacts by producing antibodies.

During the past ten years, in a private dermatologic practice, we have conducted clinical investi-

gations on the possible therapeutic value of vitamin E in the management of a number of disabling skin diseases of unknown etiology as well as several muscular disorders.

Among the diseases that were successfully controlled were a number in the autoimmune category, including scleroderma, discoid lupus erythematosus, porphyria cutanea tarda, several types of vasculitis, and polymyositis.

Since vitamin E is a physiologic stabilizer of cellular and lysosomal membranes, and since some autoimmune diseases respond to vitamin E, we suggest that a relative deficiency of vitamin E damages lysosomal membranes, thus initiating the autoimmune process.

The possibility that tocopherol (vitamin E) may be involved in the autoimmune mechanism occurred to us as the result of clinical investigations of the possible therapeutic value of vitamin E in a number of dermatoses of obscure etiology during the past ten years in our private dermatologic practice.¹ The fact that some of these treatment-resistant diseases, classified in the autoimmune category, responded to large

Dr. Ayres is Emeritus Clinical Professor of Medicine (Dermatology), University of California at Los Angeles, The Center for the Health Sciences; Dr. Mihan is Associate Clinical Professor of Medicine (Dermatology and Syphilology), University of Southern California School of Medicine.

Reprint requests to 2007 Wilshire Boulevard, Los Angeles, California 90057 (SA Jr).

doses of vitamin E suggested that a defect in the supply or utilization of, or an increased requirement for, vitamin E might be a cause. This hypothesis was further strengthened by the necessity for a continued maintenance dosage and the occurrence of relapse when the dosage was markedly reduced or discontinued, followed by favorable response when an adequate dosage was resumed.

A number of pathologic conditions involving various tissues, including the skin, have been classified as autoimmune or "autoaggressive" diseases. In these conditions, the body's immune mechanism, designed to protect the body from invasion by living microorganisms and other foreign substances, has developed antibodies that attack and destroy the body's own tissues. These antibodies can usually be demonstrated by immunofluorescent stains.

Most of the current dermatologic research on autoimmune antibodies has concerned itself with the structure of the antibodies, immunofluorescent staining, and the location of the antibodies in the tissues, but there has been little enlightenment as to why these antibodies occur. Crieep, in his comprehensive textbook,² suggested the possibility that various enzymes manufactured by lysosomes, which are intracytoplasmic organelles protected by lipoprotein membranes, may play an important role. He speculated that when these membranes are damaged, the escaping enzymes may, in some way, denature or change native protein so that it becomes antigenic, thus stimulating the production of antibodies that, in turn, react with the antigenically altered tissues. He pointed out that the anti-inflammatory effect of corticosteroids may be due in part to their stabilizing effect on lysosome membranes, thus interfering with the release of hydrolytic enzymes.

Burch³ suggested that the term "autoaggressive" is more appropriate than "autoimmune," since he believes that disturbances in immunoglobulins are more likely to represent the consequences than the primary cause of such diseases.

The mechanism of tissue damage leading to autoimmune disorders, according to Wittig et al,⁴ is largely speculative because there are no experimental counterparts. It is probable that the structural and functional integrity of the lipoprotein cell membrane is the key to the situation. Theoretically, anything that leads to the destruction of

this membrane would, in turn, be followed by destruction of the lysosome membrane, with release of hydrolytic enzymes. These might, then, cause direct damage to neighboring tissues or might damage local and remote tissues by the formation of autoimmune antibodies.

According to Tappel⁵⁻⁷ and others, lipoprotein cell membranes are unstable and tend to spontaneously combine with oxygen. This leads to lipid peroxidation, with the release of free radicals that are highly destructive to the cell and its contents, including the lysosome membrane. Destruction of this membrane could result in the release of enzymes. This lipid peroxidation probably continues to a limited extent throughout life and may account for the normal aging process. A number of experiments, both *in vivo* and *in vitro*, have demonstrated that an additional increment of alpha-tocopherol is capable of retarding the aging process.⁸⁻¹¹

Either an actual or a relative deficiency of vitamin E in cell membranes can result in cellular damage, producing various pathologic conditions. This deficiency may be due to any one of a number of factors: inadequate intake, defective intestinal absorption, defective utilization from congenital or acquired causes (including various disease states), or increased requirements (again due to congenital or acquired causes, such as an excessive intake of polyunsaturated fats, infection, and various toxic agents). Biochemical individuality, which, according to Williams,¹² means that some individuals have an inborn requirement many times the normal for certain nutrients to maintain good health, could also account for a relative deficiency of vitamin E.

Immunofluorescent antibodies, resembling those seen in pemphigus, have recently been demonstrated in skin and sera from severely burned patients,¹³ supporting the hypothesis that autoimmune antibodies can result from cellular damage—caused in this case by a physical agent, heat, rather than by a slow process of lipid peroxidation.

Among the conditions we have observed to respond favorably to vitamin E therapy are some that are generally regarded as autoimmune diseases, as indicated by the presence of immunofluorescent antibodies. These include scleroderma, discoid lupus erythematosus, several types of vasculitis, and porphyria cutanea tarda as well as polymyositis. In addition to these diseases in the

autoimmune category, we have observed excellent response to vitamin E in a number of other conditions of unknown cause, several of which are suspected of being autoimmune in nature. These include granuloma annulare, subcorneal pustular dermatosis, epidermolysis bullosa, pseudoxanthoma elasticum, yellow nail syndrome, benign chronic familial pemphigus, and post-herpes zoster neuralgia. We have had two patients with Darier's disease, and are aware of eight patients with pityriasis rubra pilaris, including one of our own, who responded to a combination of vitamins A and E, having failed to respond to vitamin A alone.

The presence of immunofluorescent antibodies in the tissues in many of the just-mentioned entities lends support to the hypothesis that the cell membranes are damaged, leading to altered integrity of lysosome membranes with the escape of enzymes that then alter tissue proteins, which in turn cause the formation of autoimmune antibodies. The presence of abnormal amounts of enzymes in two of the previously mentioned conditions—collagenase in the skin in epidermolysis bullosa, as reported by Michaelson et al,¹⁴ and marked elevation of the muscle enzyme aldolase in the serum of our patient with polymyositis,¹⁵ both of which returned to normal concomitantly with clinical response following vitamin E therapy—suggests that either an actual or a relative deficiency of vitamin E may have initiated the entire chain of events.

Voron and Tonkens,¹⁶ after reviewing the literature concerning the simultaneous occurrence of either systemic or discoid lupus erythematosus and porphyria cutanea tarda, both of which are regarded as autoimmune diseases, in nineteen patients, and reporting one of their own, concluded that this association may be related to a specific pathogenic mechanism involving photosensitivity, shared by both diseases. They suggested that "skin pathology in porphyria cutanea tarda results from the action of hydrolytic enzymes which have leaked from lysosomes ruptured by a process of lipid peroxidation initiated by photo-activated free radicals." Vitamin E, through its antioxidant action, serves both as a scavenger of free radicals and a stabilizer of cell and lysosome membranes, which may account for its therapeutic effect in some of the autoimmune diseases.

Case Reports

Porphyria Cutanea Tarda—We have successfully treated with vitamin E two middle-aged women with porphyria cutanea tarda, who had been taking estrogens for many years. Both patients had typical blebs and superficial abrasions and scars from slight trauma, especially on the dorsa of the hands and forearms, hyperpigmentation and hypertrichosis of the face, and markedly elevated urinary porphyrins. They responded almost completely to large doses of d, alpha-tocopherol acetate, 1,200 to 1,600 IU daily.

In one patient, the uroporphyrin level of 3,128 $\mu\text{g}/24$ hours (normal, less than 50 $\mu\text{g}/24$ hours) decreased to zero within a period of seven months. The urinary coproporphyrin level was within normal limits. The condition relapsed after a long absence, during which she markedly reduced her vitamin E intake and ate white bread and cereals "enriched" with inorganic iron, which counteracts vitamin E. She responded to high doses of vitamin E and avoidance of iron, and the uroporphyrin level again returned to zero.

The second patient had an even more dramatic response. Her pretreatment uroporphyrin level was 2,163 $\mu\text{g}/24$ hours and her coproporphyrin level was 7,145 $\mu\text{g}/24$ hours. Six weeks after beginning vitamin E therapy in the form of d, alpha-tocopherol acetate, 1,600 IU daily, both uroporphyrin and coproporphyrin levels registered zero, with complete clearing of the skin lesions.

Discoid Lupus Erythematosus—We have not had the opportunity of treating any patients with systemic lupus erythematosus. We have treated four patients with discoid lupus erythematosus with vitamin E in doses of 900 to 1,600 IU daily, with complete or almost complete clearing. We had previously treated two patients with doses of only 300 IU daily with no benefit, thus demonstrating the importance of adequate dosage. Burgess and Pritchard¹⁷ and Welsh¹⁸ and others had previously reported good results with vitamin E in discoid lupus erythematosus.

Scleroderma, Morphea, Raynaud's Phenomenon—A number of clinical investigations reporting a favorable effect of vitamin E in the treatment of scleroderma, as well as our own personal experiences, were reviewed in 1973.¹⁹ Our series consisted of eight patients, of whom three had systemic sclerosis, three had morphea, and two had Raynaud's phenomenon. Good to excellent results were obtained in all cases with d, alpha-tocopherol acetate, 800 to 1,200 IU daily. The most spectacular case was that of a forty-five year old man with Raynaud's phenomenon. Six ulcerated fingers, three of which were gangrenous, virtually healed eight weeks after using vitamin E both orally and topically, with no recurrence on maintenance dosage during a one-year follow-up period. Since that report, we have treated twenty additional patients,

the majority of whom have shown a favorable response, depending upon the duration of the disease and the presence or absence of irreversible tissue damage.

Vasculitis—We have successfully treated five patients with vasculitis-type eruptions, including erythema nodosum, pityriasis lichenoides et varioliformis acuta, pityriasis lichenoides chronica, and purpura annularis telangiectodes, and one patient with capillaritis of two years' duration involving the lower abdomen and right thigh, possibly of drug-induced origin. Three of the patients remained clear over a two-year observation period, except on several occasions when flares occurred after medication was discontinued. The flares subsided when vitamin E intake was resumed.

Polymyositis—As the result of a serendipitous observation, we have successfully treated with vitamin E a number of conditions involving muscle dysfunction, including nocturnal leg cramps, exercise cramps, nocturnal rectal cramps, and the restless legs syndrome. One severe case of polymyositis, which is definitely in the autoimmune category, in a seventy year old woman responded spectacularly to vitamin E after failure to respond to three immunosuppressive drugs, including azathioprine, cyclophosphamide, and prednisone. Within two months after immunosuppressive medication was discontinued and d,alpha-tocopherol acetate, 1,600 IU daily, was begun, a gastrostomy tube and permanent catheter were removed, a marked serum elevation of the muscle enzyme aldolase had returned to normal, and the patient was able to leave the hospital with full recovery except for some residual pelvic muscle weakness, probably due to irreversible muscle damage.¹⁵

Vitamin E Therapy

Vitamin E, through its antioxidant action, serves both as a scavenger of free radicals and a stabilizer of cell and lysosome membranes, which could account for its wide range of therapeutic usefulness, especially in some of the autoimmune diseases. Thomas²⁰ offered evidence indicating that the anti-inflammatory action of corticosteroids may be based on their stabilizing effect on the membranes of lysosomes. When applicable, vitamin E would appear to be a much safer stabilizer of lysosome membranes.

It must be emphasized that in the conditions mentioned herein, a disease state exists and vitamin E is administered not as a nutritional supplement, but as a therapeutic agent—to compensate for an actual or relative deficiency, however produced, of vitamin E in the involved tissues. It must be given in a potent form, in

adequate amounts, and over a long period of time, sometimes indefinitely. The most active form is d,alpha-tocopherol acetate or succinate; the dose should range between 800 and 1,600 IU or more, daily. The simultaneous ingestion of inorganic iron should be avoided, since inorganic iron combines with and inactivates vitamin E. Thus, white flour and cereals that are "enriched" with iron are interdicted. Estrogens, mineral oil, and large amounts of laxatives have been reported to have an adverse effect on the utilization of vitamin E.

The only contraindications to an initial large dose of vitamin E are marked hypertension, a badly damaged heart, or diabetes treated with insulin.²¹ In these conditions, the initial dose should not exceed 100 IU daily, which, however, can be gradually increased. We have not encountered any adverse effects. Response to vitamin E will obviously be influenced by the duration of the disease and the presence or absence of irreversible tissue damage.

Summary

In summary, our personal experience with vitamin E over a nearly ten-year period and a review of recent literature suggest the possibility that at least some of the so-called autoimmune diseases may be caused by a relative deficiency of the protective vitamin E in the lipoprotein membranes of cells and lysosomes, resulting in the destruction of these membranes with the liberation of enzymes. These in turn alter normal tissue proteins, which then stimulate the production of autoimmune antibodies. The fact that adequate amounts of vitamin E can, in some cases, reverse this process lends support to the hypothesis that vitamin E is indeed involved in the autoimmune mechanism.

In view of these circumstances, it would appear safer to employ vitamin E as a physiologic stabilizer of cellular and lysosome membranes than to prescribe various immunosuppressive agents, which not only suppress the abnormal immunologic reactions, but also destroy the body's natural defenses against infections and malignancy.

We do not wish to imply that the usefulness of vitamin E is restricted to autoimmune disorders. There are also many disease states of unknown etiology, in which the autoimmune mechanism may not be involved, that are benefited by large doses of alpha-tocopherol. In addi-

tion to its antioxidant function in protecting cell membranes, vitamin E paradoxically also facilitates oxygen utilization in normal metabolic processes²² and may participate in other physiologic activities in conjunction with certain enzymes and trace elements.

It is also probable that some diseases classified in the autoimmune category may not respond to vitamin E because of other associated pathologic conditions or irreversible tissue damage.

References

1. Ayres S Jr, Mihan R: Vitamin E and dermatology. *Cutis* 16: 1017, 1975.
2. Criepp LH: Clinical Immunology and Allergy, 2nd edition, p 3. New York, Grune & Stratton, 1969.
3. Burch PRJ: Autoimmune or autoaggressive disease? A new approach to etiology. *Dermatol Digest* 10: 30, 1971.
4. Wittig HJ, Welton WA, Burrell R: A Primer on Immunologic Disorders, p 79. Springfield, Illinois, Charles C Thomas, 1970.
5. Tappel AL: Vitamin E as the biological antioxidant. *Vitam Horm* 20: 493, 1962.
6. Tappel AL: Where old age begins. *Nutr Today* 2: 2, 1967.
7. Tappel AL: Lipid peroxidation damage to cell components. *Fed Proc* 32: 1870, 1973.
8. Harman D: Free radical theory of aging: effect of free radical inhibitors on the life-span of male LAF mice: second experiment. *Gerontologist* 8 (pt 2): 13, 1968.
9. Tuchweber B, Gabbiani B, Selye H: Effect of vitamin E and methyltestosterone upon the progeria-like syndrome produced by dihydrotachysterol. *Am J Clin Nutr* 13: 238, 1963.
10. Einarson L: Cellular structure in the central nervous system of vitamin E deficient monkeys. In, *Aging Around the World: Biological Aspects of Aging: Proceedings of the Fifth Congress of the International Association of Gerontology*, p 131. New York, Columbia University Press, 1962.
11. Packer L, Smith JR: Vitamin E extends cell's lifespan. *Med World News* 15: 47, 1974.
12. Williams RJ: Nutrition Against Disease. New York, Pittman, 1971.
13. Ablin RJ, Milgrom F, Kano K, et al: Pemphigus-like antibodies in patients with skin burns. *Vox Sang* 16: 73, 1969.
14. Michaelson JD, Schmidt JO, Dresden MH, et al: Vitamin E treatment of epidermolysis bullosa. *Arch Dermatol* 109: 67, 1974.
15. Killeen RNF, Ayres S Jr, Mihan R: Polymyositis: response to vitamin E: a case report. *South Med J* 69: 1372, 1976.
16. Voron DA, Tonkens SW: Lupus erythematosus coexisting with porphyria cutanea tarda: lysosomal photoreactivity as a common denominator. *Cutis* 15: 69, 1975.
17. Burgess JF, Pritchard JE: Tocopherol (Vitamin E) treatment of lupus erythematosus: preliminary report. *Arch Dermatol Syphilol* 57: 953, 1948.
18. Welsh AL: Lupus erythematosus: treatment by combined massive amounts of pantothenic acid and vitamin E. *Arch Dermatol Syphilol* 70: 181, 1954.
19. Ayres S Jr, Mihan R, Levan NE, et al: Raynaud's phenomenon, scleroderma and calcinosis cutis: response to vitamin E. *Cutis* 11: 54, 1973.
20. Thomas L: The role of lysosomes in tissue injury. In, *The Inflammatory Process* (Zweifach BW, Grant L, McCluskey RT, eds), chapt 12, p 450. New York, Academic Press, 1965.
21. Shute WE: Precautions in the use of alpha-tocopherol in the treatment of hypertensive heart disease. *Ann NY Acad Sci* 52: 354, 1949.
22. Schwarz K: A possible site of action for vitamin E in intermediary metabolism. *Am J Clin Nutr* 9 (pt 2, suppl): 71, 1961.

1977 Scholarship Awards of the American Society for Dermatologic Surgery

The American Society for Dermatologic Surgery began an annual scholarship program in 1975 to emphasize the scope and importance of surgery in dermatology. Eligible candidates were dermatology residents with at least six months of training remaining prior to the surgical elective; full-time faculty in approved dermatology residency programs; or those involved in a dermatology training program in some other capacity. The surgical elective proposed by the candidate was to broaden his or her surgical experience and to acquire skills in new procedures.

The American Society for Dermatologic Surgery is pleased to announce the winners of its 1977 scholarship program. They are:

1. A. Razzaque Ahmed, MD, of UCLA;
2. Jeffrey H. Binstock, MD, of Albany Medical College;
3. Robert A. Breedlove, MD, of the University of Tennessee;
4. Ronald S. Davis, MD, of Tulane University;
5. Scott C. Duncan, MD, of the Mayo Clinic;
6. Richard I. Ecker, MD, of the Mayo Clinic;
7. Ira Gouterman, MD, of Yale University;
8. William Hendricks, MD, of Case Western Reserve;
9. Hal B. Ridgway, MD, of the University of Wisconsin;
10. Bryan C. Schultz, MD, of Northwestern University;
11. G. Thomas Spigel, MD, of the Mayo Clinic;
12. Jon K. Stern, MD, of Baylor College of Medicine; and
13. Herschel E. Stoller, MD, of the University of Nebraska.

Our Society offers its congratulations to these scholarship winners.